

Image-Based Visual Servoing of Quadrotors to Arbitrary Flight Targets

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Abstract—Visual servoing of Unmanned Aerial Vehicles (UAVs) has achieved satisfactory performance in fixed and planar motion targets. Due to highly coupled system dynamics and the sensitivity of the target image to aircraft attitude, the problem for chasing free-flying targets remains challenging. In this paper, a vision-based algorithm is designed for controlling an UAV while tracking an intruder flying arbitrarily in 3D space. Image-based visual servoing is used to design controllers that depend directly on errors in image plane. Specifically, a virtual camera approach is adopted to decouple the UAV dynamics by compensating the pitch and yaw, and an improved image error term is proposed to reduce the impact of the UAV rotation on error signals in the process of tracking, thus a simplified control design is achieved, and the stability of the visual servo system is guaranteed. Comparison and ablation experiments in both simulated and real environments are provided to verify the effectiveness of the proposed method.

Index Terms—Aerial systems, perception and autonomy, visual servoing.

I. INTRODUCTION

WITH frequent reports of small aircraft breaking into no-fly zones (e.g. airports, school campuses and military bases) in recent years, the problem of how to monitor and intercept these intruding aerial targets is becoming a research hotspot. One viable solution is to employ an autonomous UAV to approach and capture the intruders [1]. To accomplish this task, some of onboard sensors are often necessary, such as global positioning system (GPS) receivers, inertial measurement units (IMU), LiDAR and cameras. As a growing trend, cameras are used by researchers as the primary sensor for studying the autonomous flight of UAVs [2], [3], [4], [5], [6]. The reason

for this can be summarized as the cameras can provide useful information in almost any environmental and climatic conditions, and at the same time they are light in mass and low in cost.

Visual servo aims to control a robot to the desired position based on visual perception [7]. Among different methods, image-based visual servoing (IBVS) is more attractive because it computes the control signal directly from image features on the image plane and is less sensitive to errors generated during system modeling and camera calibration [8], [9]. However, the design of the IBVS controller of the UAV is considered challenging since the system dynamics are underactuated, nonlinear and strong-coupling [10].

There have been a number of works about IBVS control of UAVs in recent years. In [10], [11] and [13], [14], [15], [16], [17], [18], the IBVS schemes based on spherical image moments and perspective image moments have been proposed, respectively. These image features have been extracted to obtain decoupled dynamic equations and facilitated control design. In [12], a virtual coordinate frame has been defined in IBVS. The image roll and pitch can be compensated in this coordinate system, and the desired decoupling of the vehicle dynamics can be obtained. Similarly, the idea of introducing virtual image plane can be found in [13], where image features are reprojected to the virtual image plane to guarantee effective trajectories in Cartesian space. But the design of the controller requires the assumption that the linear velocity of the vehicle is known. Afterwards, without linear velocity measurements, a translational controller of an UAV has been proposed for tracking a planar motion target in the presence of constant external disturbances [14]. This work has been modified in [15] by defining auxiliary systems and robustifying terms in controller to handle time-varying external disturbances. By restricting the thrust, reference roll and pitch to a compact set, an input saturation visual servoing law has been provided in [16]. Then, in [17], a nonlinear velocity observer has been presented based on the image moment features with a backstepping approach to develop the IBVS controller, which can be applied to track a visual target with three point features in a tilted plane. Furthermore, the virtual camera servoing scheme has been extended to an arbitrarily oriented planar target in [18]. However, all the works mentioned above consider the tracking IBVS problem with the assumption that the target is fixed or in planar motion. It is definitely more difficult in tracking a 3D space free-motion target, because the error signals used for controller design from image coordinates of the target are easily

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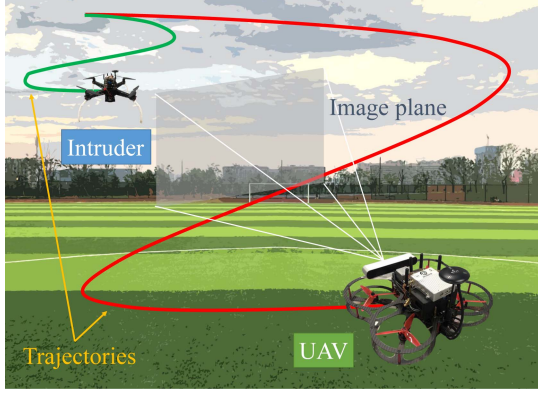


Fig. 1. The overall research scenario of the paper. The intruder flies arbitrarily in the air while the UAV can automatically track it. The tracking algorithm based on IBVS runs on the on-board computer.

impacted by the rotation of the UAV during tracking, especially pitch and yaw motions.

Meanwhile, some learning-based methods are used in visual servoing and UAV control in recent years. In [19], Q-learning has been adopted to select the servo gains adaptively, and a fuzzy controller has been proposed for adjusting the learning rate. An end-to-end learning based framework has been introduced for visual servoing tasks in [20]. However, the special environment for parameter learning in reinforcement learning limits the application of the method to practical scenarios. At the same time, some data driven approaches are proposed for UAVs to learn control strategies. In [21], the imitation learning techniques have been used to train the UAV controller based on a set of human pilot demonstrations. And in [22], a convolutional neural network has been proposed to drive a UAV to fly in cities by imitating manned vehicles. However, since the data collected from the real world is noisy, the additional data pre-processing techniques and learning time are required by the above data driven methods.

In this paper, we design and analyze a vision-based control scheme for an UAV, which is able to track the intruder flying arbitrarily in three-dimensional space (see Fig. 1). The aircraft is equipped with an IMU and a fixed forward-looking camera. A virtual camera method is utilized for the projection of the intruder's center point onto a virtual image plane, which is always perpendicular to the ground plane. A IBVS controller, designed based on the improved error signal, is considered that stabilizes the full dynamics of the UAV. As a result, an efficient and continuous process of tracking an aerial target is achieved. In simulation and real-world experiments, the intruder is tested to fly at three different speeds along three representative trajectories in linear, planar and 3D spaces, respectively.

In a closely related work [23], an IBVS algorithm is also proposed to intercept intruders by drones. A simplified rigid body dynamics model is used to facilitate the controller design, which limit the maneuverability of the UAVs. In summary, the reprojection image features used and the full dynamics considered in this paper are the major differences between their work and ours. Due to the highly nonlinear, strong-coupled and

underactuated properties of the full dynamics model, effective controller design becomes more challenging. By defining the visual feature errors in the virtual image plane, the impact of aircraft rotation during tracking flight on error signals is attenuated. And by adopting the full dynamics model in IBVS, the high speed attitude control is achieved [24].

The contributions of this paper are summarized as follows:

- To the best of authors' knowledge, this is the first work that applies the virtual camera approach to tracking IBVS of arbitrary motion targets in 3D space.
- A new image error term is proposed for the stabilization of the dynamics of the UAV. Compared to the classical methods [25], [26], the medians and the interquartile ranges of the image errors are reduced by 22.37% and 35.10%, respectively.
- In comparison with the most related work [23], this paper considers the full dynamics model of the UAV. Comparative experiments are provided to verify the efficiency of the proposed algorithm.

The rest of this paper is organized into four sections. The fundamental model equations for a quadrotor and visual servo are presented in Section II. An IBVS control design for the translational and rotational motions is provided in Section III. The details of the simulations and experiments are shown in Section IV. Finally, the conclusions and the directions for future researches are discussed in Section V.

II. SYSTEM MODEL

The quadrotor dynamics model and the visual servo model based on the virtual camera approach are described in this section, respectively.

A. Quadrotor Model

In this paper, the quadrotor UAV is considered to have a rigid structure, i.e., it is treated as a rigid body. Let $\mathcal{E} = \{X_e, Y_e, Z_e\}$ denote the world frame in North-East-Down coordinates, and $\mathcal{B} = \{X_b, Y_b, Z_b\}$ denotes the body-fixed frame which orients forward, right and down direction of the vehicle. Consider a quadrotor UAV modeled as [14]

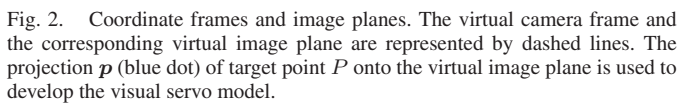
$$\dot{\xi} = V \quad (1)$$

$$m\dot{V} = mg e_z - f R e_z \quad (2)$$

$$\dot{\eta} = T\omega \quad (3)$$

$$J\dot{\omega} = \tau - sk(\omega)J\omega \quad (4)$$

where $\xi = [x, y, z]^T \in \mathbb{R}^3$ and $V \in \mathbb{R}^3$ represent the position and the linear velocity of the quadrotor in $\mathcal{E}$, respectively. The aircraft's mass is denoted by m , and the gravitational acceleration is denoted by g . Let f denote the thrust force and $R: \mathcal{B} \rightarrow \mathcal{E}$ denotes a rotation matrix. Let $e_z = [0, 0, 1]^T$ be the unit vector in $\mathcal{E}$. The exogenous torque is denoted by $\tau \in \mathbb{R}^3$, and the inertia matrix of the body-fixed frame is denoted by J , respectively. Let $sk(\cdot)$ be the skew-symmetric matrix such that $sk(a)b := a \times b$, where $a, b \in \mathbb{R}^3$, and $\times$ denotes the cross-product.


$$\mathbf{T} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix}$$

B. Visual Servo Model

Assume that the intruder is a stationary target point P . The coordinates of P in $\mathcal{E}$ and $\mathcal{V}$ are represented by ${}^e\mathbf{p} = [{}^ex, {}^ey, {}^ez]^T$ and ${}^v\mathbf{p} = [{}^vx, {}^vy, {}^vz]^T$, respectively. Their conversion equation can be written as

$${}^v p = R_{\phi}^T ({}^e p - O_v)$$

$$\mathbf{R}_\phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}.$$
$$\begin{aligned}
{}^v\dot{\mathbf{p}} &= \dot{\mathbf{R}}_\phi^T({}^e\mathbf{p} - \mathbf{O}_v) - \mathbf{R}_\phi^T\dot{\mathbf{O}}_v \\
&= -\mathbf{s}\mathbf{k}(\dot{\phi}\mathbf{e}_x)\mathbf{R}_\phi^T({}^e\mathbf{p} - \mathbf{O}_v) - \mathbf{R}_\phi^T\dot{\mathbf{O}}_v \\
&= -\mathbf{s}\mathbf{k}(\dot{\phi}\mathbf{e}_x){}^v\mathbf{p} - \mathbf{v}
\end{aligned} \tag{5}$$

The coordinate of the target point P in the virtual image plane is denoted by $\mathbf{p}$. Then, the perspective projection equation in $\mathcal{V}$ becomes

$$\mathbf{p} = \begin{bmatrix} v_u \\ v_n \end{bmatrix} = \frac{\lambda}{v_x} \begin{bmatrix} v_y \\ v_z \end{bmatrix}.$$

$$\dot{p} = J_v \dot{r}_v \quad (6)$$
$$\mathbf{J}_v = \begin{bmatrix} {}^v u / {}^v x & -\lambda / {}^v x & 0 & -{}^v n \\ {}^v n / {}^v x & 0 & -\lambda / {}^v x & {}^v u \end{bmatrix}$$

It is known that the distance between the intruder and UAV in the x direction is very important to accomplish surveillance and interception missions through tracking. And if we can estimate this distance value in real time, the situation that the quadrotor first approaches and then moves away from the target can be avoided. Hence, based on the idea of monocular camera distance estimation, we design Algorithm 1.

III. CONTROLLER DESIGN AND STABILITY ANALYSIS

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Algorithm 1: Distance Estimation And Stop Condition Judgment Algorithm.

Input: Camera horizontal resolution N_e , Maximum number of horizontal pixels of the target image N , Constant parameters d_e and k_d

Output: Distance d , Forward tracking velocity $^v v_x$

```

1: while  $N > 0$  do
2:   if  $d \leftarrow (N_e/N)d_e, \bar{d} \leftarrow k_d d_e$   $\triangleright$ Distance estimation
3:   while  $^v v_x > 0$ 
4:     if  $d > \bar{d}$  then
5:        $^v v_x \leftarrow ^v v_x$   $\triangleright$ Continue tracking
6:     else
7:        $^v v_x \leftarrow 0$   $\triangleright$ Stop approaching
8:     end if
9:   end while
10: end while

```

quadrotor and intruder converges to zero, the control objectives are considered to be achieved. During the tracking process, it is assumed that the speed of the quadrotor is always faster than that of the intruder, otherwise the tracking cannot be completed.

A. Control of the Translational Motion

In this paper, the control law is first designed and then we calculate the UAV thrust. The error signal in the virtual image plane is defined as

$$\mathbf{e}_i = \mathbf{p} - \mathbf{p}_d$$

where the desired coordinates of P are denoted as $\mathbf{p}_d = [0, 0]^T$ (to center the image patch of the target in the virtual image plane). Then, the time-derivative of $\mathbf{e}_i$ satisfies

$$\dot{\mathbf{e}}_i = \dot{\mathbf{p}}. \quad (7)$$

The derivative of $\mathbf{p}$ is a function of time t instead of $\mathbf{e}_i$, thus system (7) is non-autonomous. To prove that $\mathbf{e}_i$ converges to zero, consider the following quadratic Lyapunov function

$$V_i = \frac{1}{2} \mathbf{e}_i^T \mathbf{e}_i.$$

The derivative of V_i along the trajectories of (7) gives

$$\dot{V}_i = \frac{^v v_x ^v u^2}{^v v_x} - \frac{\lambda ^v v_y ^v u}{^v v_x} + \frac{^v v_x ^v n^2}{^v v_x} - \frac{\lambda ^v v_z ^v n}{^v v_x}.$$

We aim to design a controller and then find conditions to guarantee that $\mathbf{e}_i$ converges to zero. First, by considering $^v v_y$ and $^v v_z$ as the control input, the controller is designed as $^v v_y = k_1 ^v u, ^v v_z = k_2 ^v n$. We then have

$$\dot{V}_i = -\frac{^v u^2(k_1 \lambda - ^v v_x)}{^v v_x} - \frac{^v n^2(k_2 \lambda - ^v v_x)}{^v v_x}.$$

Next, we uncover conditions for uniform stability of $\mathbf{e}_i$ [Definition 4.4, 27]. According to [Theorem 4.9, 27], it suffices to show that $\dot{V}_i \leq -W(\mathbf{e}_i)$, where $W(\mathbf{e}_i)$ is a positive definite function of $\mathbf{e}_i$. For simplicity, let $W(\mathbf{e}_i) = 1/2 \mathbf{e}_i^T \mathbf{e}_i$. (Note that this is not the only choice of W , and other forms can also be designed.)

For $\dot{V}_i \leq -W(\mathbf{e}_i)$, i. e.,

$$\dot{V}_i = -\frac{^v u^2(k_1 \lambda - ^v v_x)}{^v v_x} - \frac{^v n^2(k_2 \lambda - ^v v_x)}{^v v_x} \leq -\frac{1}{2} \mathbf{e}_i^T \mathbf{e}_i$$

to hold, it suffices to have

$$\begin{cases} \frac{^v x}{2} + ^v v_x \leq k_1 \lambda, \\ \frac{^v x}{2} + ^v v_x \leq k_2 \lambda, \end{cases}$$

which serves as a sufficient condition that ensures uniform stability of $\mathbf{e}_i$.

The intuition of the above condition is as follows. Since k_1 , k_2 , and λ are all positive, to achieve tracking task, there is a trade-off between flying speed and the distance to the intruder of the quadrotor. Specifically, to track the target, if the quadrotor is far from the intruder, then the speed of the quadrotor cannot be too fast. This is reasonable since a high speed too far from the target may cause the problem of feature loss [28].

Next, we calculate the thrust. The position dynamics equation (2) in the direction of Z_e axis can be written as

$$^e \dot{v}_z = g - \frac{f}{m} \cos \phi \cos \theta. \quad (8)$$

According to the relationship between the world frame $\mathcal{E}$ and the virtual camera frame $\mathcal{V}$, $^e v_z$ can be expressed as

$$^e v_z = ^v v_z \cos \phi. \quad (9)$$

By using (8) and the time-derivative of (9), consider $k_2 ^v n$ as the desired velocity, the total thrust f is given as

$$f = \frac{m}{\cos \phi \cos \theta} [^v v_z \sin \phi + k_3 (^v v_z - k_2 ^v n) \cos \phi + g] \quad (10)$$

where $k_2, k_3 > 0$ are the proper control gains for $^v v_z$ tracking the desired velocity.

B. Control of the Roll Rotation

The quadrotor dynamics function (2) in $\mathcal{V}$ can be written as

$$\dot{\mathbf{v}} = -\mathbf{s}\mathbf{k}(\dot{\phi} \mathbf{e}_x) \mathbf{v} - \frac{\mathbf{R}_{\theta\psi} \mathbf{e}_z f}{m} + g \mathbf{e}_z \quad (11)$$

where $\mathbf{R}_{\theta\psi} = \mathbf{R}_\psi \mathbf{R}_\theta$. Utilizing $\dot{\mathbf{R}} = \mathbf{R}\mathbf{s}\mathbf{k}(\omega)$, the time-derivative of $\mathbf{R}_{\theta\psi}$ is given by [13]

$$\dot{\mathbf{R}}_{\theta\psi} = \mathbf{R}_{\theta\psi} \mathbf{s}\mathbf{k}(\omega) - \mathbf{s}\mathbf{k}(\dot{\phi} \mathbf{e}_x) \mathbf{R}_{\theta\psi}. \quad (12)$$

By combining (10), (11) and (12), ω_2 can be obtained. To control the image patch of the intruder close to the center of the camera image, the desired yaw angle, $\psi_d = 0$, and ω_3 can be calculated as

$$\omega_3 = k_\psi (\psi - \psi_d)$$

where k_ψ is a positive parameter.

For roll controller design, the error function is defined as

$$e_\phi = \phi - \phi_d$$

where $\phi_d = 0$ is the desired value of the roll angle, because the tracking process is expected to be smooth and stable. According to (3), the rate of change of ϕ can be calculated as

$$\dot{\phi} = \omega_1 + \sin \phi \tan \theta \omega_2 + \cos \phi \tan \theta \omega_3. \quad (13)$$

By choosing ω_1 as the control input, the roll rotation controller is designed as

$$\omega_1 = -\sin \phi \tan \theta \omega_2 - \cos \phi \tan \theta \omega_3 - k_\phi e_\phi \quad (14)$$

where $k_\phi > 0$ is an adjustable constant.

Substituting (14) into (13), the error dynamics can be developed as

$$\dot{e}_\phi = -k_\phi e_\phi,$$

which implies e_ϕ will converge to zero exponentially.

IV. SIMULATIONS AND EXPERIMENTS

A variety of simulation and real-world experiments are conducted to demonstrate the effectiveness of the proposed controllers in this section. The different trajectories of intruder movement are used to verify the tracking performance of the UAV.

A. Simulation Results

Both the Robot Operating System (ROS) [29] and the PX4 autopilot [30] are adopted in all simulation experiments. ROS, a widely used software framework, is able to control low-level devices and communicate messages between processes (nodes) in the development of robot applications. Besides, the PX4 can be applied to make individual nodes as local ROS nodes. Hence, the autopilot and the operating system have good compatibility.

The simulation models and environment are built based on the Gazebo [31]. The UAV and the intruder considered in simulation tests are two quadrotors. The length, width and height of both of them are equal to 0.47 m, 0.47 m and 0.11 m, respectively. The other model parameters of the UAV are selected as $m = 1.55$ kg, $g = 9.8$ m/s², and $\mathbf{J} = \text{diag}\{0.029125, 0.029125, 0.055225\}$ kgm². The forward-looking camera can take images in 640×480 resolution at 20fps, and its field of view is 1.047 rad.

For all three simulation experiments, the intruders are considered to move freely in linear, planar and 3D spaces. The initial positions of the intruders are (0, 0, 0) m in the world frame, and the UAVs start at 20.0 m directly behind the intruders. The control gains are selected as $k_1 = 6$, $k_2 = 8$, $k_3 = 3$, $k_\psi = 2$, and $k_\phi = 3.5$. After the simulation starts, the intruder and the UAV take off and hover at an altitude of 10.0 m based on the position mode (see Fig. 3). Then, their flight modes are switched to the offboard mode. The intruder starts flying, and its trajectory in this mode is determined by the representative given position equation. At the same time, the images of the intruder are captured by the onboard camera of the UAV, and the UAV starts tracking flight.

In the first simulation, the intruder is designed to fly along a straight line at a speed of 4.0 m/s. The forward flight speed of the UAV is 5.0 m/s. In the time period of 0-37 s, the quadrotor UAV is armed and run preflight sensor/estimator checks. Then, it starts to take off at the time of $t = 37$ s and run the tracking algorithm at $t = 44$ s. The tracking error values for the linear velocities of the UAV and intruder in the world frame $\mathcal{E}$ are shown in Fig. 4(a). This can be seen in Fig. 4(b) illustrating the time evolution of the distances between the UAV and intruder.

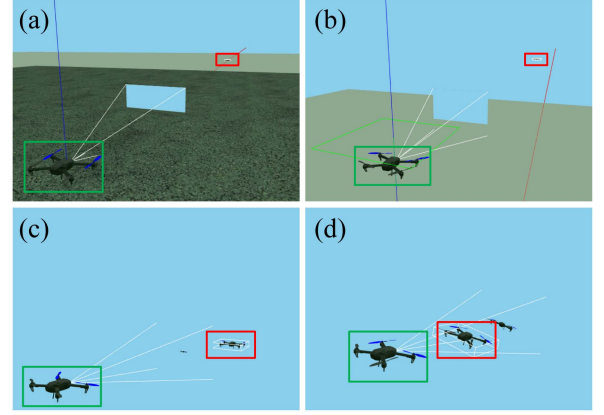


Fig. 3. Snapshots of tracking process in simulation environment: (a) takeoff, (b) hovering, (c) tracking and (d) completion. The UAV and the intruder are marked with green and red bounding boxes, respectively.

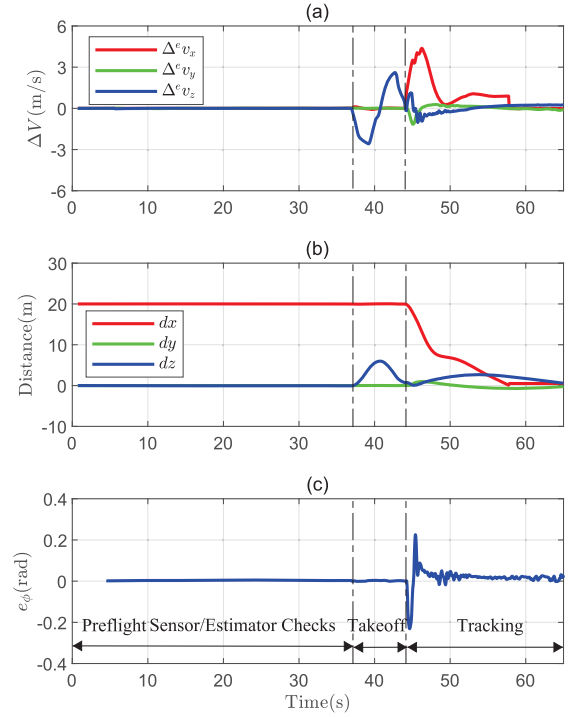


Fig. 4. Simulation 1: the intruder flies along a straight line. (a) Velocity tracking error, (b) Distances between UAV and intruder, and (c) roll angle error.

As a result, the convergence to zero of the distances of X_e , Y_e and Z_e axes lets us know that the UAV tracks and approaches the intruder. In addition, Fig. 4(c) represent the roll angle error signal. Note that the inevitable jitters of the state and error values reflected in the figures are caused by mode switching. Before $t = 4$ s, the UAV is disarmed, and the IMU is not working. Hence there is no data on e_ϕ being demonstrated in the figure.

In the second simulation, the intruder is assumed to fly along a sine line with amplitude of 5.0 m. The speeds of the intruder and the UAV are 4.5 m/s and 5.0 m/s, respectively. The results obtained from this test are shown in Fig. 5. Fig. 7(a) depicts the trajectories of the UAV and intruder from the start to the

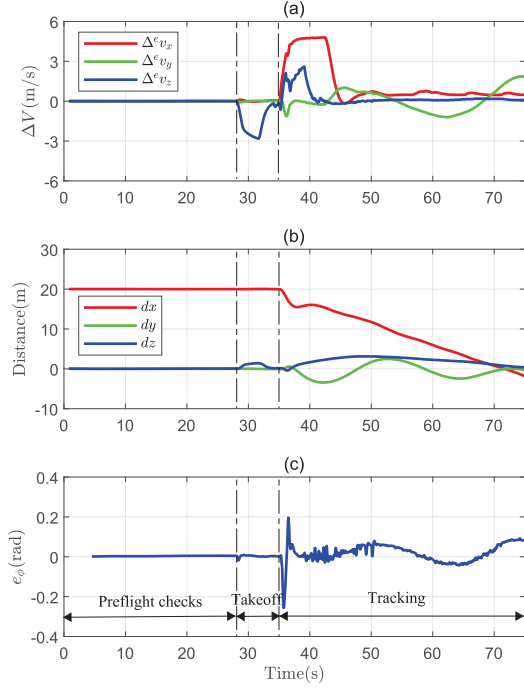


Fig. 5. Simulation 2: the intruder flies along a sine line. (a) Velocity tracking error, (b) Distances between UAV and intruder, and (c) roll angle error.

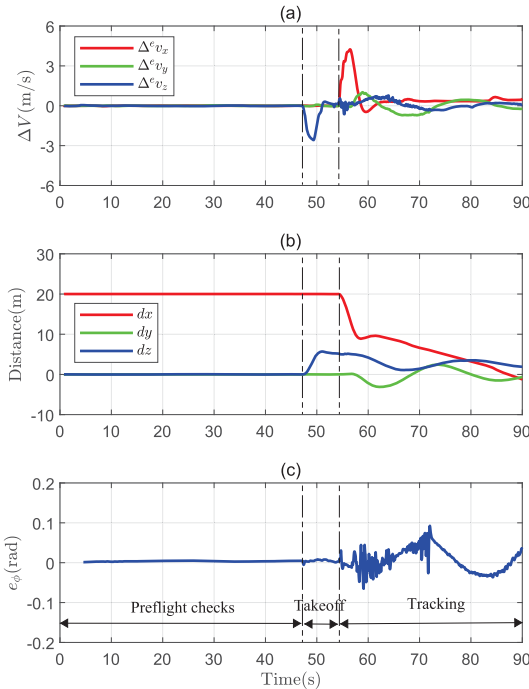


Fig. 6. Simulation 3: the intruder flies along a helix line. (a) Velocity tracking error, (b) Distances between UAV and intruder, and (c) roll angle error.

completion of the tracking mission, which indicates that the tracking controller is effective for planar free-motion targets.

In the last simulation, with a speed of 4.6 m/s, the intruder flies along a helix line whose radius and pitch are 5.0 m and 120.0 m. The forward speed of the UAV is still 5.0 m/s. Fig. 7(b) and Fig. 6 illustrate that the proposed approach still performs well in

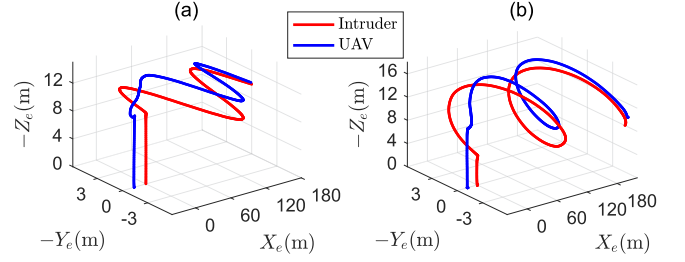


Fig. 7. (a) Simulation 2 and (b) simulation 3: trajectories of UAV and intruder.

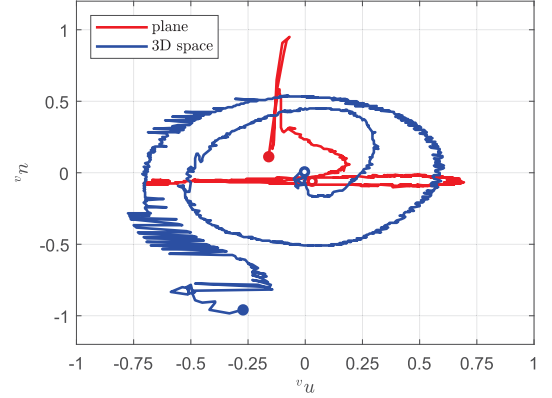


Fig. 8. Simulation 2 - 3: trajectories of intruder in virtual image plane. The results of intruder's planar and 3D motions are represented by red and blue lines, respectively. The filled dots show the positions at the beginning, and the circles stand for ending positions.

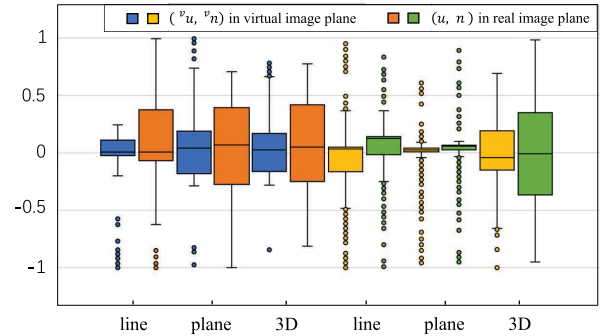


Fig. 9. Ablation experiments: medians and variation ranges of image errors in the virtual image plane (ours) and real image plane.

3D space. Finally, the trajectories of intruder points in the virtual image plane obtained from the data of the last two simulation experiments are shown in Fig. 8. When v_u and v_n converge to zero, the image patch of the intruder reaches the center of the image plane, and the aircraft tracks and approaches the target.

Also, three ablation experiments are designed in the simulation environment where the intruders are considered to fly freely in linear, planar and 3D spaces, respectively. The median, quartiles, bounds and outliers of the image error data are presented in Fig. 9. It can be seen that the medians from the improved image error term in the virtual image plane are closer to zero than that in the real image plane, and the variation ranges of the error

TABLE I
QUANTITATIVE METRICS FOR IMPROVED IMAGE ERROR TERM AND CLASSICAL
ERROR TERM

		(v_u, v_n)	(u, n)
Linear motion	median	(0.006, 0.033)	(0.006, 0.125)
	IQR	(0.136, 0.215)	(0.444, 0.158)
Planar motion	median	(0.041, 0.025)	(0.069, 0.050)
	IQR	(0.370, 0.033)	(0.669, 0.042)
3D motion	median	(0.025, -0.042)	(0.050, -0.008)
	IQR	(0.331, 0.342)	(0.669, 0.717)
Improvement	median	22.37%	
	IQR	35.10%	

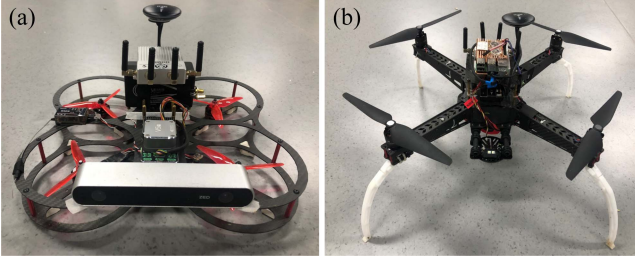


Fig. 10. (a) Quadrotor UAV and (b) intruder. The UAV is equipped with a ZED camera, a video transmitter and a NVIDIA TX2. Pixhawk autopilot and GPS receiver are needed by both the UAV and intruder for flight and localization outdoors.

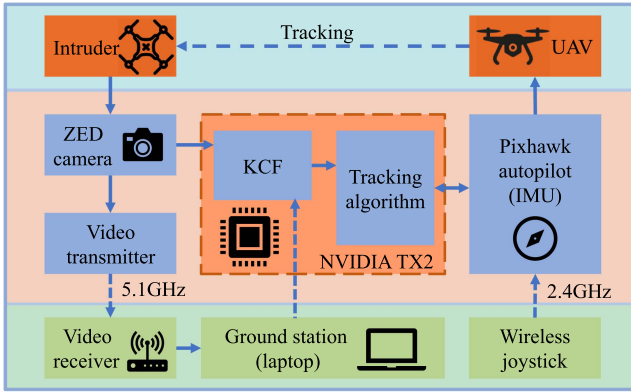


Fig. 11. Illustration of data transmission in experiment.

signals in our method are much smaller in the all three ablation experiments.

From Table I, we can obtain that the medians of the image errors in our approach are reduced by 22.37% on average compared to the previous approach, and the Interquartile Ranges (IQR: difference between upper and lower quartiles) are reduced by 35.10% on average.

B. Experimental Tests

The two quadrotors assembled for the real-world experiments are shown in Fig. 10. A ZED stereo camera installed on the front of the UAV is used to capture video frames. Behind the camera, a Pixhawk autopilot with a built-in IMU is loaded to estimate angular velocities and the Euler angles and generate motor control signals. An NVIDIA Jetson TX2 is equipped on the vehicle to process the image data from onboard camera and

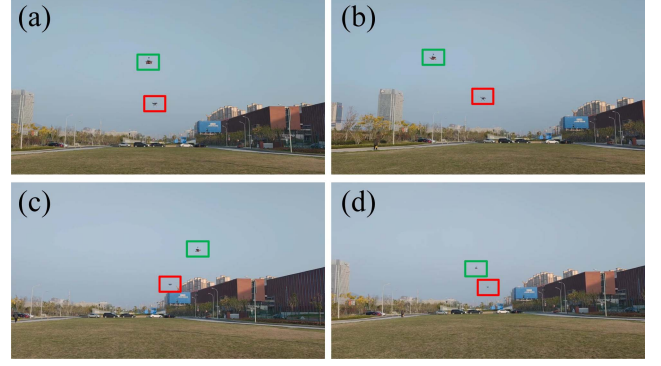


Fig. 12. Snapshots of tracking process in real-world environment: (a) start tracking, maneuver to the (b) left and (c) right, and (d) forward flight. The UAV and the intruder are marked with green and red bounding boxes, respectively.

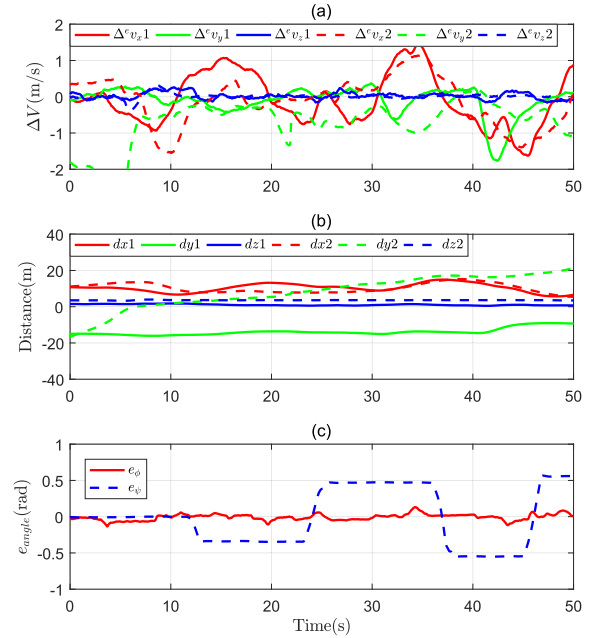


Fig. 13. Comparative experiment: note that our method is marked with solid lines and the method proposed in [23] marks dashed lines. (a) Velocity tracking error, (b) distances between UAV and intruder, and (c) roll and yaw angle errors.

compute the control commands (thrust and angular velocity) based on the proposed algorithm. In addition, the image data can be sent to a laptop for selection of the intruder by a 5.1 GHz video transmitter located on the top of the UAV.

In Fig. 11, the entire setup of the experiment is exhibited. The kernelized correlation filter (KCF) [32] is utilized as the visual tracker to track the intruder in video frames after selection. In other words, the image patch containing the intruder needs to be manually selected in the first frame, then the KCF algorithm can track the target in the subsequent frames and send the coordinates of the target to the controllers. The wireless joystick is used to switch modes and control the UAV in unexpected situations.

In the comparison experiments, the same initial conditions for the tracking UAV and the intruder are set to compare the attitude error and the tracking performance of our method with the

method proposed in [23]. The illustrations of the experimental procedure are displayed in Fig. 12.

The results from the experiments are presented in Fig. 13. The evolution of the velocity tracking errors with time are shown in Fig. 13(a). It can be seen that the error variation ranges are similar in the x and z directions, but the tracking error of the approach [23] jitters more dramatically in the y direction than our approach. This is because the controller in [23] is based on a simplified dynamics model which requires the small angle assumption ($\phi, \theta \approx 0$) as a prerequisite. As a result, the maneuverability of the tracking UAV is limited. The relative positions of the UAV and intruder are illustrated in Fig. 13(b). After 50 seconds, the distance error based on our method converges to 10.30 m, which is smaller than the error in their method (20.83 m). Fig. 13(c) shows the attitude angle errors. It can be observed that the error of our method is much smaller, and as the distance decreases, the attitude error of the tracking UAV in [23] gradually increases.

V. CONCLUSION

In this paper, to solve the problem of tracking and monitoring the arbitrary flight target in 3D space, a vision-based control scheme for UAVs is proposed. A virtual camera approach is shown efficient to provide appropriate UAV dynamics for the image-based visual servoing design. The controllers for translation and rotation of the aircraft are proposed based on the improved image error term. Furthermore, three UAV models and two real quadrotors are adapted to conduct simulations and experiment, respectively, and all the results are shown to indicate effective performance with the proposed approach. Due to the limited field of view of the camera, feature loss may still occur when the intruder is too far away or faster than the tracking UAV [28]. Therefore, our future work will focus on a visual surveillance system with multiple UAVs and other types of robots.

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